

simTIM

A Temporal Cybersecurity Simulator

Noah Niclas Behrisch

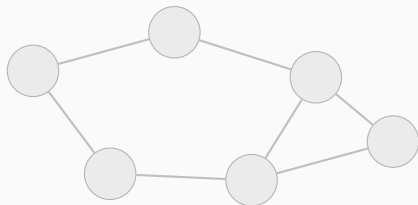
University of Halle-Wittenberg, Germany

ARES 2026

Zoltán Ádám Mann

University of Münster, Germany

Why model cybersecurity?

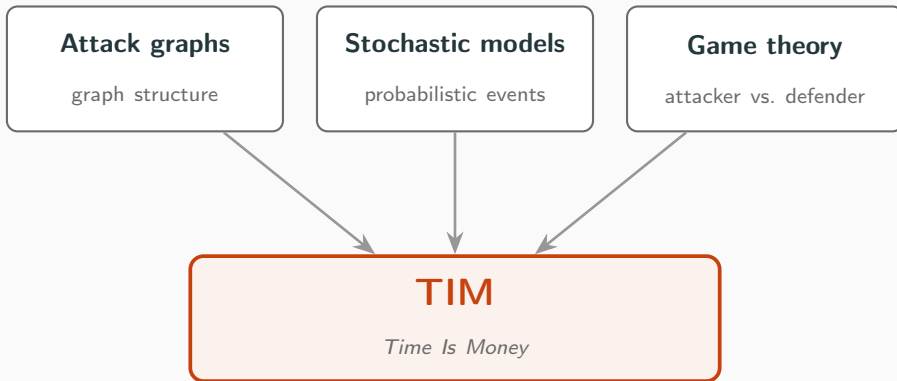


Which servers are vulnerable and need hardening?

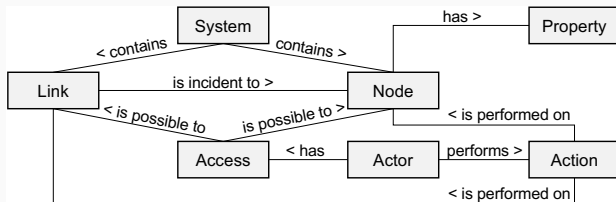
How should we change the network topology?

Simulations provide **what-if analysis**.

TIM: three families of security models



What TIM models



Nodes and links

servers, routers, connections

Attackers: maximise gain

Defenders: minimise damage

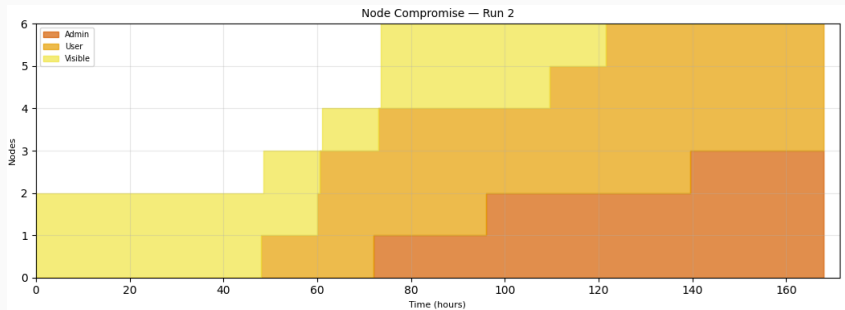
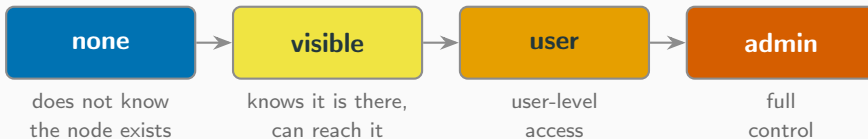
Actions

$$a = (\varphi_a, \psi_a, c_a, d_a, p_a)$$

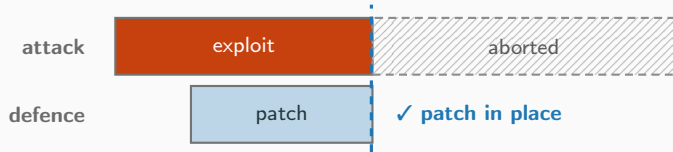
precondition, postcondition,

cost, duration, success probability

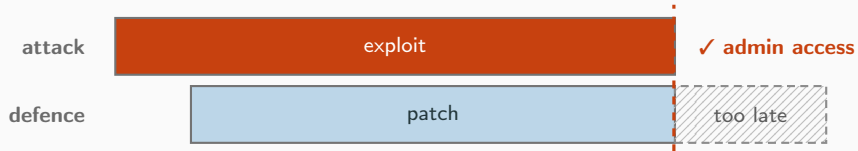
Access levels



Defence wins



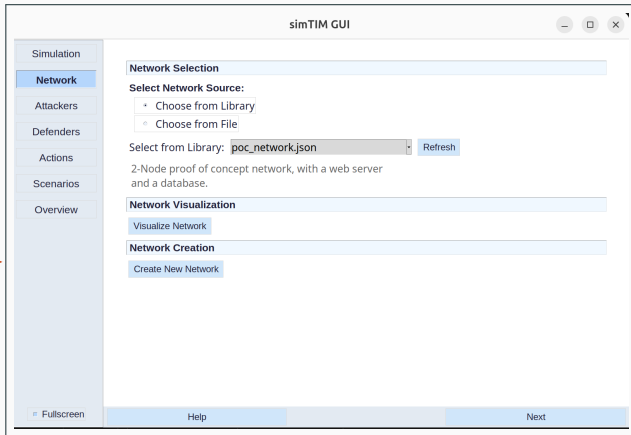
Attack wins



simTIM: a full implementation of TIM

```
n1 = Node("web server")
n1.set("framework", ...)
n2 = Node("database")
...
attacker.do(a_tapestry, n1)
```

proof of concept
a *subset* of TIM



- ▶ graphical interface
- ▶ configurable simulations
- ▶ Monte Carlo runs
- ▶ visualisation

Walkthrough

Simulation

Network

Attackers

Defenders

Actions

Scenarios

Overview

☐ Fullscreen

Simulation Parameters:

Simulation Runs:

Simulation Duration: days hours

Detection Engine:

Detection Engine:

Help

Next

Simulation

Network

Attackers

Defenders

Actions

Scenarios

Overview

☐ Fullscreen

Network Selection

Select Network Source:

* Choose from Library

☐ Choose from File

Select from Library:

demo_network.json

Refresh

Custom network with 6 nodes and 6 links

Network Visualization

Visualize Network

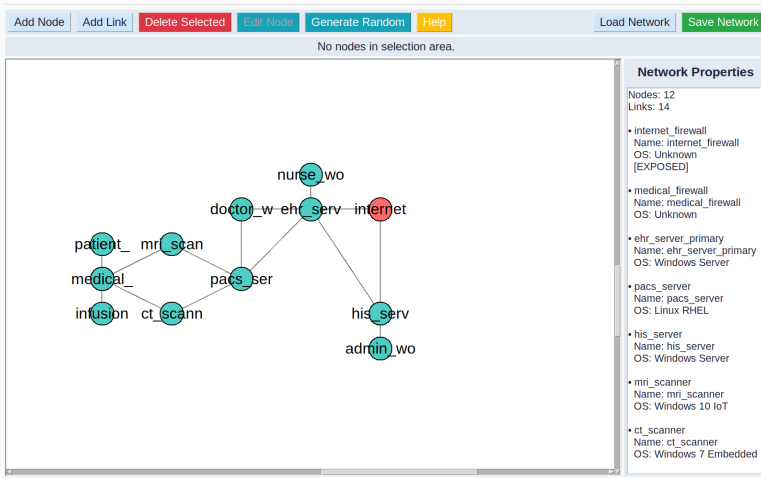
Network Creation

Create New Network

Help

Next

Network Creator



Simulation

Network

Attackers

Defenders

Actions

Scenarios

Overview

Attacker Strategies:
Available: escalation, greedy, random

ID	Strategy	Capacity	∞	Budget (\$)	Actions
Attacker 1	escalation		∞	100000	Remove

Add Attacker

☐ Fullscreen

Help

Next

Simulation

Network

Attackers

Defenders

Actions

Scenarios

Overview

Defender Strategies:

Available: balanced, monitoring, proactive, reactive

ID	Strategy	Capacity	Budget (\$)	∞	Actions
Defender 1	balanced	2		∞	Remove

Add Defender

Fullscreen

Help

Next

The screenshot displays the 'Actions Configuration' window. On the left is a sidebar with navigation links: Simulation, Network, Attackers, Defenders, **Actions** (highlighted), Scenarios, and Overview. The main area is titled 'Actions Configuration' with a subtitle 'Enable/disable actions for simulation.' It is divided into two columns: 'Attack Actions' and 'Defense Actions'. Each column contains a list of actions with checkboxes. At the bottom of the main area are four buttons: 'Enable All Attacks', 'Disable All Attacks', 'Enable All Defenses', and 'Disable All Defenses'. The bottom of the window features a 'Fullscreen' button on the left and 'Help' and 'Next' buttons on the right.

Simulation

Network

Attackers

Defenders

Actions

Scenarios

Overview

Actions Configuration

Enable/disable actions for simulation.

Attack Actions

- ☒ Application Layer Protocol: Web Protocols
- ☒ Command and Scripting Interpreter: PowerShell
- ☒ Data Encrypted for Impact
- ☒ Data from Local System
- ☒ Exfiltration Over C2 Channel
- ☒ Exploit Public-Facing Application
- ☒ Exploitation for Privilege Escalation (Linux)

Defense Actions

- ☐ Comprehensive Security Audit
- ☐ Data Backup Restore
- ☒ Decoy File
- ☒ Emergency Security Patch
- ☒ Firewall Rule Update
- ☐ Honeypot Deployment
- ☒ Incident Response

Enable All Attacks Disable All Attacks Enable All Defenses Disable All Defenses

☐ Fullscreen Help Next

Simulation

Network

Attackers

Defenders

Actions

Scenarios

Overview

Variable Scenarios

▼ Enable Scenario Comparison

Parameter to vary: defense_duration

When disabled, simulations use defense action durations from JSON files.
When enabled, compare different defense action duration scenarios.

#1	Defense Duration (hours):	1	Runs:	20	×
#2	Defense Duration (hours):	5	Runs:	20	×
#3	Defense Duration (hours):	20	Runs:	20	×

+ Add Scenario Clear All

✓ 3 defense duration(s): 1.0h, 5.0h, 20.0h | Total runs: 60

Fullscreen

Help

Next

Simulation

Network

Attackers

Defenders

Actions

Scenarios

Overview

SIMULATION CONFIGURATION
=====

Simulation Parameters:

- Mode: DEFENSE DURATION COMPARISON
- Scenarios: 3
- Total Runs: 60
- Time per run: 168 hours
- Detection Engine: early_weighted

Scenario Details:

1. Defense Duration: 1.0h → 20 runs
2. Defense Duration: 5.0h → 20 runs
3. Defense Duration: 20.0h → 20 runs

Network Configuration:

- File: demo_network.json
- Path: networks/library/demo_network.json

Attackers (1):

1. attacker1
 - Strategy: escalation
 - Capacity: ∞
 - Budget: \$inf

Defenders (1):

☐ Fullscreen

Help

Results

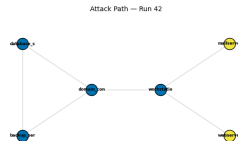
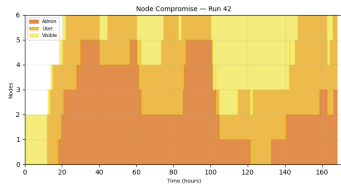
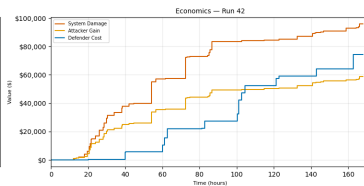
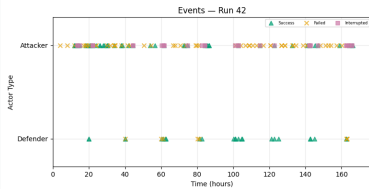
Start Simulation

16 / 24

Dashboard

Dashboard Event History attacker1 defender1 Events Timeline Money Timeline Nodes Timeline Attack Path Statistical Analysis

Select Run: Run 42



Event Log

[00:24] attacker1 detected Exploit Public-Facing Application → webserver

Event History

Dashboard Event History attacker1 defender1 Events Timeline Money Timeline Nodes Timeline Attack Path Statistical Analysis

Select Run: Run 42

Run 42 Summary: Total Events: 212 | Successful: 76 | Detections: 37

[00:24]  attacker1 detected Exploit Public-Facing Application → webserver

[03:00]  attacker1 detected Phishing → mailserver

[04:00]  attacker1 failed Exploit Public-Facing Application → webserver

[08:00]  attacker1 failed Exploit Public-Facing Application → webserver

[09:32]  attacker1 detected Exploit Public-Facing Application → webserver

[12:00]  attacker1 failed Phishing → webserver

[12:00]  attacker1 succeeded Phishing → mailserver

[12:00]  attacker1 succeeded Exploit Public-Facing Application → webserver

[12:30]  attacker1 succeeded Network Service Discovery (Link) → mailserver<->workstation_win

[13:30]  attacker1 interrupted Phishing → workstation_win

[13:30]  attacker1 interrupted Remote Desktop Protocol → mailserver<->workstation_win

[13:30]  attacker1 succeeded Remote Desktop Protocol → webserver<->workstation_win

[14:00]  attacker1 succeeded Network Service Discovery (Link) → workstation_win<->domain_controller

[14:39]  attacker1 detected Exploitation for Privilege Escalation (Windows) → workstation_win

[14:46]  attacker1 detected Remote Desktop Protocol → domain_controller

[15:00]  attacker1 interrupted Phishing → domain_controller

[15:00]  attacker1 succeeded Remote Desktop Protocol → workstation_win<->domain_controller

[17:00]  attacker1 failed SSH → domain_controller<->database_server

[18:00]  attacker1 failed Exploitation for Privilege Escalation (Linux) → mailserver

[18:00]  attacker1 succeeded Exploitation for Privilege Escalation (Linux) → webserver

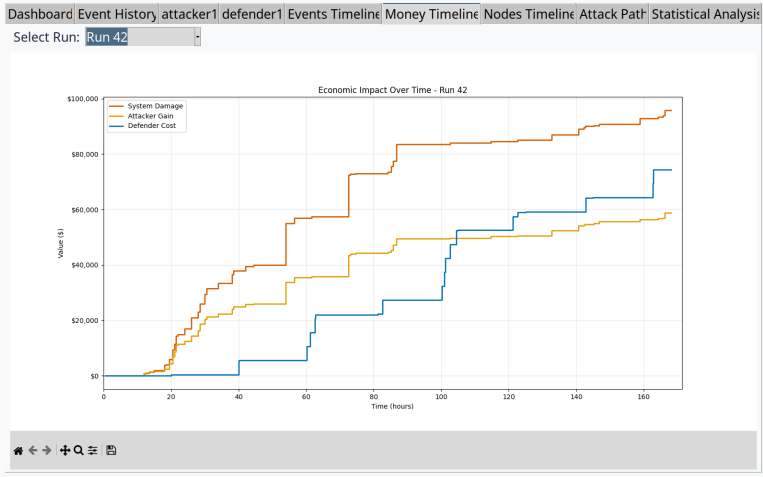
[18:30]  attacker1 failed Remote Desktop Protocol → domain_controller<->backup_server

[18:30]  attacker1 succeeded Network Service Discovery (Link) → domain_controller<->backup_server

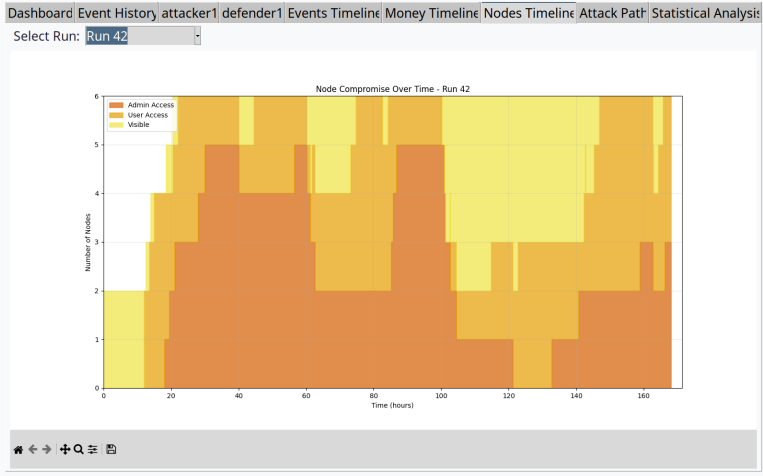
[19:30]  attacker1 succeeded Exploitation for Privilege Escalation (Windows) → workstation_win

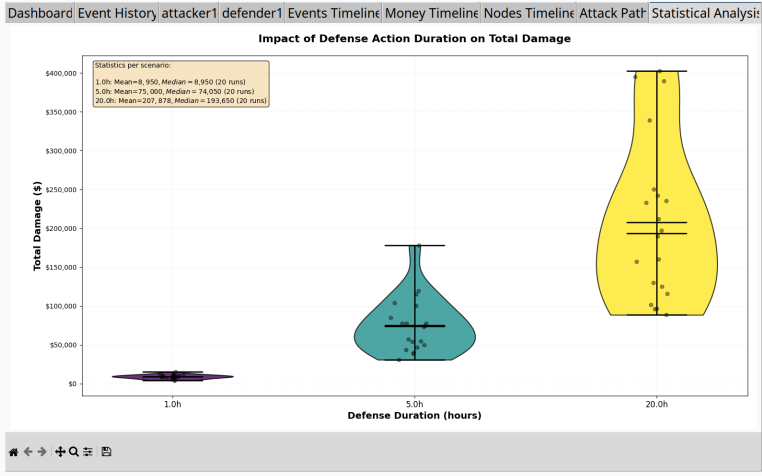
[20:00]  defender1 succeeded Technical Detection System → database_server

Money Timeline



Nodes Timeline







github.com/noahbehrisch/simTIM

Apache 2.0 · **actively developed**

```
make install  
make run
```

Four extension points:

- ▶ networks (JSON)
- ▶ actions (JSON)
- ▶ **strategies** (one class)
- ▶ detection engines (one class)

Limitations

- Simplistic strategies
- Only a subset of real actions
- No real-world data

Future work

- More strategies and actions
- More sophisticated simulation behaviour
- More data, more realism